Preliminary results master thesis

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1. Setup: outcomes, windows, and seasonal control

(Two) reforms. The Spanish wholesale electricity market shifted from a 60-minute to a 15-minute market time unit in two steps: **ID15** (2025-03-19, intraday auctions and continuous market) and **DA15** (2025-10-01, day-ahead). After each reform there are four periods per clock-hour; before, one. A third granularity reform, **ISP15** (2024-12-11), moved REE's settlement period[1] from 60 to 15 minutes but left the OMIE contract grid unchanged. The *operación reforzada* regulatory stance (REE's post-blackout reinforced operation, continuous from 2025-04-28) is a continuous confound that forces CCGT post-dayahead inclusion; tight pre/post windows around each reform date hold its level constant within each arm. Figure 1 summarises the full timeline, including other reforms that are not directly relevant for our research question but are important for context.

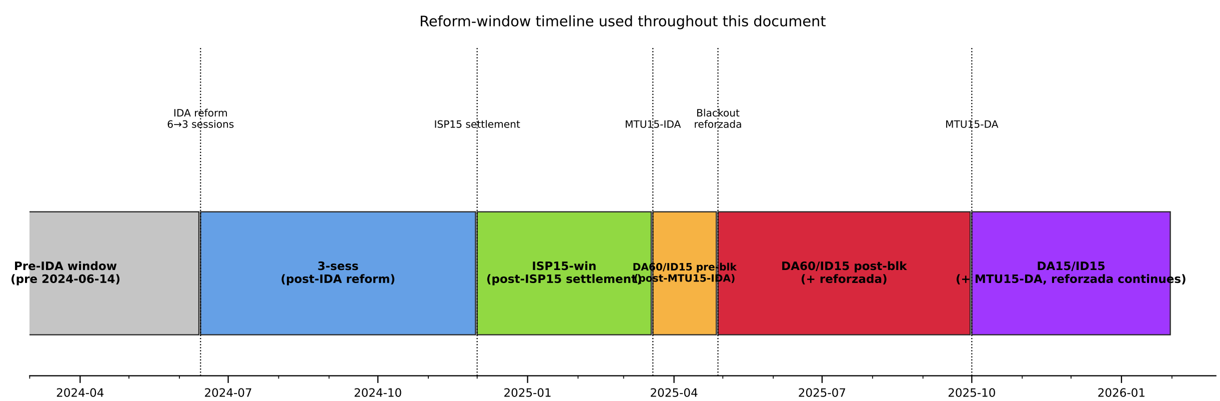

Figure 1: Full reform-window timeline. Five named windows partition the post-2024-06-14 sample; the pre-IDA window is the baseline regime. Dotted vertical lines mark the five reform dates that produce the window boundaries.

Variables. Each (unit u, date d, period p) sell offer is a tranche ladder $\{(p_k, \Delta q_k)\}_{k=1}^{K_{u,d,p}}$ sorted by ascending price, where p_k is the bid price (EUR/MWh) and Δq_k is the *incremental* MW added to the unit's step supply function at p_k[2]. Offers carry one of three structural types: *simple* (a bare price–quantity ladder), *minimum-income condition* (MIC), or *block* (all-or-nothing across periods).

[1] There are penalizations for the units when they deviate from their scheduled production.

[2] Following OMIE *Potencia* field in *ficheros OMIE v1.37* §5.1.4.2. Δq_k is the marginal MW at tranche k; the unit's total offered capacity in the period is $\sum_k \Delta q_k$, and the cumulative supply at any price p is $S(p) = \sum_{k:\, p_k \leq p} \Delta q_k$.

Most technologies use simple bids, but CCGT is the outlier with the bulk of its offers in MIC or block form (§ 4, § 5(d))[3]. Let $\mathrm{MCP}_{d,p}$ be the OMIE clearing price *of the market in which the offer was submitted* and h a bandwidth that isolates the bids in contention: tranches close enough to the clearing price to plausibly compete in the *competing tier* of the offer (CCGT's two-tier structure makes this distinction explicit, § 7). We anchor h to the *contemporary contested margin*: h is computed as $p_{90} - p_{50}$ of the MCP distribution *in the analysis window of the comparison, market by market.* The resulting eight bandwidths (one per reform $\times$ {real, placebo} $\times$ {DA, IDA}) are summarised in the table below; bandwidth-robustness across wider h choices is reported in § 5(e).

Window	h in DA (EUR/MWh)	h in IDA (EUR/MWh)
ID15 real	50	62
ID15 placebo (2024)	45	46
DA15 real	50	58
DA15 placebo (2024)	45	49

Table 1: Bandwidths h used to define the in-band region. h is set to $p_{90} - p_{50}$ of the MCP distribution in each (window, market). Window-and-market-specific by construction. Bandwidth robustness in § 5(e).

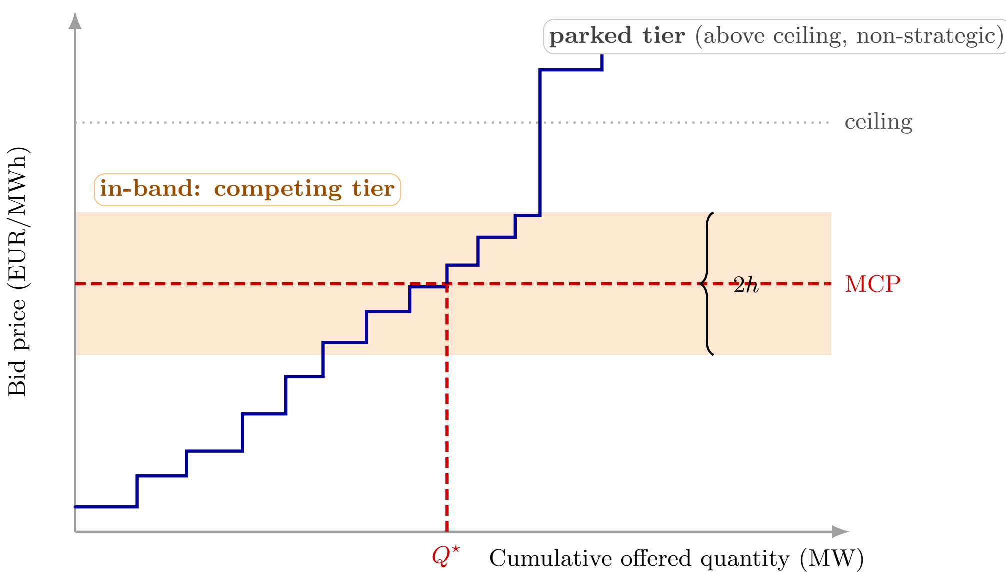

Figure 2: Bandwidth h and the in-band region. The supply curve is the sorted sell-bid ladder for one quarter and one market. The market clears where supply meets demand, fixing the MCP. Bid-shape outcomes (σ_p, N_{eff}) are computed only on tranches in the in-band region $[\mathrm{MCP} - h,\ \mathrm{MCP} + h]$ — the competing tier where strategic action lives. The parked tier above the clearing-price ceiling is a non-strategic placeholder (§ 7) and is excluded by construction.

Now define the in-band set, MW weights, and in-band weighted mean bid price:

$$\mathcal{B}_{u,d,p} \;=\; \{k : |p_k - \mathrm{MCP}_{d,p}| \le h\}, \qquad w_k \;=\; \frac{\Delta q_k}{\sum_{j \in \mathcal{B}_{u,d,p}} \Delta q_j}, \qquad \bar{p}_{u,d,p} \;=\; \sum_{k \in \mathcal{B}} w_k\, p_k.$$

[3]Approximate sell-offer shares by structural type (DA15/ID15): wind, solar, hydro and pump-storage ∼100% simple; nuclear 64% simple, 36% MIC; CCGT 18% simple, 50% MIC, 32% block.

Per-curve (in-band, MW-weighted; one value per (unit, date, period)):

$$\sigma_{p,\,u,d,p} \;=\; \sqrt{\sum_{k\in\mathcal{B}_{u,d,p}} w_k\,(p_k - \bar{p}_{u,d,p})^2}, \qquad N_{\text{eff},\,u,d,p} \;=\; \Big(\sum_{k\in\mathcal{B}_{u,d,p}} w_k^2\Big)^{-1}.$$

σ_p = price spread of the bid ladder within a single quarter; N_{eff} = effective tranche count of that quarter (1 = a single block, large = many equal-sized tranches).

Aggregate: $\text{MCP}_{d,p}$ is the OMIE clearing price (EUR/MWh); $Q_{d,p} = \sum_{u,k} q^{\text{cleared}}_{u,d,p,k}$ is the system DA cleared quantity per (date, period)[4].

In-band quantity primitive. For each (unit u, date d, period p, IDA session s when applicable) curve, the in-band MW is

$$m_{u,d,p[,s]} \;=\; \sum_{k\in\mathcal{B}_{u,d,p[,s]}} \Delta q_k,$$

the total tranche MW falling in the $\pm h$ band around the MCP of that curve. The two quantity outcomes below are both built from m.

Within-hour dispersion (post-only, used to calibrate the critical/flat partition in § 2): for a clock-hour with four quarters $q = 1,\ldots,4$ post-MTU15, let $\bar{m} = \frac{1}{4}\sum_q m_q$ (averaging the per-quarter m values). For each (unit, date, clock-hour) we compute the standard deviation across the four quarters of the clock-hour for each of the five outcomes[5]:

$$\begin{aligned} D_{\text{price}} = \text{sd}_q(\bar{p}_q), \quad D_{\text{qty}} = \text{sd}_q(m_q)/\overline{m}, \quad \text{SD}_{\text{price}} = \text{sd}_q(\text{MCP}_q),\\ D_\sigma = \text{sd}_q(\sigma_{p,q}), \quad D_N = \text{sd}_q(N_{\text{eff},q}). \end{aligned}$$

D_{price}, D_{qty}, SD_{price} are across-quarter shifts in bid LEVEL, bid VOLUME, and system clearing prices; D_σ, D_N are across-quarter shifts in ladder SPREAD and TRANCHE COUNT. All five are zero under the MTU60-equivalent strategy (identical bids in every quarter), so any positive value post-MTU15 is a direct trace of within-hour discrimination.

In-band offered energy (aggregate, per (date, market, tech, hour-class)). The same primitive $m_{u,d,p[,s]}$ aggregated across tranches (already inside m), units of tech t, periods p inside an hour, hours h in a class $\mathcal{X}$, and IDA sessions s when applicable, then converted to energy and normalised by class size:

$$E^{\text{band}}_{t,m,d,\mathcal{X}} \;=\; \frac{1}{|\mathcal{X}|}\sum_{h\in\mathcal{X}}\sum_{p\in h}\sum_{u\in t,[s]} m_{u,d,p[,s]}\cdot\tau_p,$$

where τ_p is the period length in hours (1 pre-MTU15, 1/4 post-MTU15). The $m\cdot\tau_p$ product converts MW to MWh, making the sum invariant to the period count (MTU15 quadrupling does not inflate it mechanically). The $1/|\mathcal{X}|$ normalisation expresses the result as MWh per class-hour,

[4]The OMIE assigned-power field is per-period instantaneous power (MW), invariant to period length, so one 15-min quarter and one 60-min hour at the same MW are directly comparable as rates. Aggregating MW across the four quarters of a clock-hour gives the stock of energy delivered in that clock-hour, $\text{MWh}_h = \sum_{q\in h}\text{MW}_q\,\Delta t_q$. Aggregate-quantity coefficients below are reported as MWh per clock-hour (numerically equal to the per-period mean MW because the integration is over 1 h, but the unit is energy, not power).

[5]We ommit the u, h subscripts for readability.

directly comparable across classes of different sizes. (Note that D_{qty} above is already a cross-window-comparable normalised quantity — a coefficient of variation in m across the four quarters — so it needs no further rescaling; E^{band} needs both τ_p and $|\mathcal{X}|$ to be cross-window comparable because it sums rather than ratios.)

Cross-market wedge. Per-clock-hour day-ahead/intraday price gap and its within-day spread:

$$\Delta_{d,h} \;=\; \text{MCP}^{\text{DA}}_{d,h} - \text{MCP}^{\text{IDA}}_{d,h}, \qquad \Delta^{\text{SD}}_{d} \;=\; \text{sd}_h(\Delta_{d,h}).$$

$\Delta_{d,h}$ is cross-market disagreement at clock-hour h; Δ^{SD}_{d} is how much that disagreement varies across the 24 clock-hours of day d. Both $\text{MCP}^{\text{DA}}_{d,h}$ and $\text{MCP}^{\text{IDA}}_{d,h}$ are computed at the clock-hour level by averaging *within each market* over whatever sub-hour structure that market has on date d[6]: DA collapses the four 15-min quarters post-DA15 (one observation pre-DA15); IDA collapses the three auction sessions and (post-ID15) the four within-hour quarters. The wedge then compares two equally-aggregated clock-hour means and is therefore directly comparable across regimes: it does not mechanically widen just because IDA gains a quarter dimension. Under asymmetric granularity (ID15: DA 60-min, IDA 15-min), each clock-hour hosts four IDA quarter-prices against a single DA price, so $\Delta_{d,h}$ can move further where within-hour residual demand actually varies — this is a behavioural channel, not a mechanical one.

Seasonality. Spanish electricity has strong intra-year seasonality. It bites on the level outcomes — clearing prices and per-tech cleared energy — which co-move with heating demand in winter, cooling demand in summer, and the hydrological cycle. Bid-shape outcomes (σ_p, N_{eff}) are MCP-centered and MW-weighted on the in-band ladder, so price-level seasonality does not transmit.

2. The critical/flat partition: where granular pricing has economic content

Intuition. Granular pricing only has economic content where within-hour residual demand[7] actually varies; if a clock-hour is internally flat, the four quarters look the same and there is nothing to discriminate on. We measure within-hour variation by $\sigma_{\text{within}}(h,d) = \text{sd}_q[\text{load} - \text{wind} - \text{solar}]_{q,h,d}$ on 15-min residual-demand data[8] and partition clock-hours by median σ_{within}:

- **Critical** $\mathcal{C} = \mathcal{C}_M \cup \mathcal{C}_E$ with $\mathcal{C}_M = \{5,\ldots,8\}$ (*morning ramp*) and $\mathcal{C}_E = \{16,\ldots,22\}$ (*evening ramp*); $\sigma_{\text{within}} \in [522, 1303]$ MW overall. The two sub-classes have different seasonal patterns (sunrise/sunset timing shifts the start of solar generation across the class boundaries), and bid-side analyses below treat them separately so each sub-class's seasonality is modelled on its own pattern.
- **Flat** $\mathcal{F} = \{1,2,3\}$: overnight; $\sigma_{\text{within}} \in [336, 357]$ MW.
- **Midday** $\{11,\ldots,14\}$ (solar-marginal, $\sigma_{\text{within}} \in [468, 502]$ MW): excluded from main specs; kept for falsification (§ 5(a)).

[6] Not sure about this, maybe I should exploit all the granularity!

[7] Spanish-market convention calls this the *hueco térmico* (thermal gap) — the load that thermal and other dispatchable generation must cover after subtracting wind and solar.

[8] ENTSO-E Transparency Platform: A65 (actual total load) and the wind/solar component of A75 (actual generation per type, PSR codes B16, B18, B19); window 2023-01-01 to 2026-05-14.

Formal calibration of the partition (post-only). Let $D_{d,h}$ stand for any one of the five within-hour dispersion outcomes (D_{price}, D_{qty}, SD_{price}, D_σ, D_N) on date d, clock-hour h. Under MTU60 the within-hour quarter dimension does not exist, so $D_{d,h}$ is mechanically zero pre-reform; we use the post-MTU15 sample to *describe* where within-hour dispersion lives across hour-classes. A date-FE regression is a convenient way to summarise the critical−flat differential:

$$D_{d,h} \;=\; \delta_d \;+\; \theta\,\mathbf{1}\{h \in \mathcal{C}\} \;+\; \varepsilon,$$

fit per outcome. θ is the post-MTU15 critical−flat dispersion gap. This is a **descriptive contrast, not a causal estimate**.

Outcome	**Subsample**	**ID15**	**DA15**	**crit/flat ratio (IDA, DA)**
D_{price} (EUR/MWh)	per unit (pooled)	0.39^{***} (0.05)	1.35^{***} (0.11)	4.1×, 7.4×
D_{qty} (CV)	per unit (pooled)	0.024^{***} (0.003)	0.033^{***} (0.002)	4.3×, 4.9×
SD_{price} (EUR/MWh)	system	4.55^{***} (0.40)	3.97^{***} (0.37)	2.8×, 2.6×
D_σ (EUR/MWh)	CCGT	−0.24 (0.35)	0.57^{***} (0.10)	–, 5.9×
	Hydro	0.48^{***} (0.07)	1.22^{***} (0.16)	2.2×, 3.1×
	Hydro pump	0.16^{***} (0.03)	0.66^{***} (0.15)	3.7×, 10.6×
	Wind	−0.003 (0.005)	0.04^{***} (0.02)	1.0×, 2.3×

Table 2: Post-MTU15 critical−flat within-hour dispersion gap, by outcome. *SE clustered by date*; $^{***}p < 0.01$. Bid-level rows (D_{price}, D_{qty}, SD_{price}) are bandwidth-free. The per-tech D_σ rows are computed at the new p_{90} baseline (§ 1): ID15 column uses post-ID15 IDA data with h=62; DA15 column uses post-DA15 DA data with h=50. *Reading.* Under DA15, D_σ is positive and significant for all three dispatchable techs in DA — CCGT 0.57, hydro 1.22, pump-storage 0.66 — with the largest critical/flat ratio for CCGT (5.9×). Under ID15, hydro and pump-storage widen their IDA within-hour ladders (0.48 and 0.16), while CCGT IDA D_σ is not distinguishable from zero — consistent with combined-cycle gas mostly competing in DA rather than IDA at this stage. Wind stays small everywhere, the price-taking baseline.

3. Identification: DiD for bid shape, BSTS for prices and quantities

u unit; d date; p period; h clock-hour; s IDA session; T_r reform date. Treatment status $D_{d,h} \equiv \mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h \in \mathcal{C}\}$; sample restricted to $h \in \mathcal{C} \cup \mathcal{F}$ for the DiD.

3.A. Spec A: Bayesian Structural Time Series for prices, cleared energy, and wedge SD

Idea. A Bayesian Structural Time Series (BSTS) model (Brodersen et al., 2015), as applied to the EPEX 15-min reform by Märkle-Hußet al. (2020), fits a flexible state-space decomposition of the response on the pre-reform window and extrapolates forward to construct a counterfactual: what the outcome *would* have looked like in the post-window absent the reform. The treatment effect is the gap between observed and counterfactual. The strength over other methods is that BSTS handles complex seasonality and trend *internally*, useful here since the pre-windows are short.

Model. The daily response decomposes additively into trend, weekly cycle, exogenous covariates, and noise:

$$y_t \;=\; \underbrace{\mu_t}_{\text{trend}} + \underbrace{\gamma_t}_{\text{weekly cycle}} + \underbrace{X_t'\beta}_{\text{exogenous covariates}} + \underbrace{\varepsilon_t}_{\text{noise}}.$$

μ_t is a random walk with stochastic slope (a slowly evolving local level), γ_t a 7-day cycle, and $X_t = (\text{wind GWh}, \text{solar GWh}, \text{gas price EUR})$ exogenous covariates that absorb merit-order shifts driven by renewable supply and fuel prices (a spike-and-slab prior on β shrinks unused regressors). Demand is excluded to avoid simultaneity.

Outcomes y_t. DA or IDA daily-average clearing prices, per-tech daily cleared GWh, within-day wedge SD Δ_d^{SD} overall and per hour-class $\Delta_d^{\text{SD},\mathcal{X}} = \text{sd}_{h\in\mathcal{X}}(\Delta_{d,h})$ for $\mathcal{X} \in \{\mathcal{C}, \mathcal{M}, \mathcal{F}\}$.

Estimand. Conditional on X_t, the post-cutover counterfactual equals the pre-trained extrapolation $\hat{y}_t(0)$. The average treatment effect on the post-window is

$$\bar{\tau} \;=\; \tfrac{1}{|\text{post}|} \textstyle\sum_{t\geq T_r} \big(y_t - \hat{y}_t(0)\big),$$

reported together with its 95% credible interval from the MCMC posterior produced by the `CausalImpact` R package.

Windows hold the relevant regulatory regime constant across pre and post (Figure 3): DA15 pre starts at the reforzada onset; ID15 pre starts at the post-IDA-reform 3-session architecture and stops at the blackout.

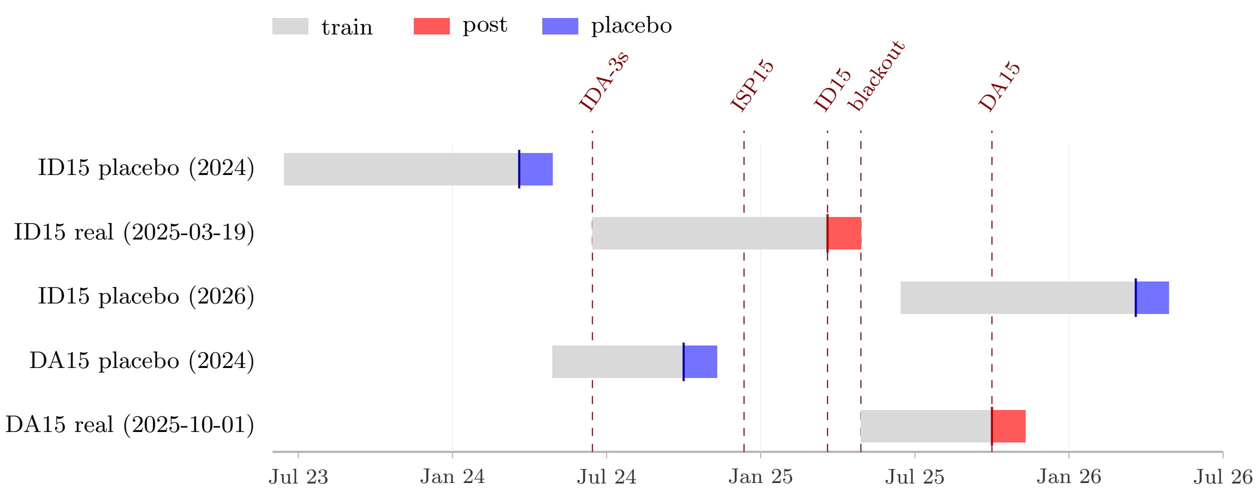

Figure 3: Spec A windows. Each row is one BSTS run. Grey: pre-window (model training). Red: post-window after the real reform cutover. Blue: post-window after a same-calendar placebo cutover (one year before or after the real reform). Pre-windows are restricted so that the relevant regulatory regime is held constant across pre and post: DA15 pre starts at the reforzada onset (2025-04-28); ID15 pre starts at the post-IDA-reform 3-session architecture (2024-06-14) and stops at the blackout (2025-04-28). The independent 2026 placebo for ID15 doubles the falsifier — a calendar-symmetric counterpart on the other side of the real reform.

Placebo cross-check and the joint 95% credible interval. The same BSTS is re-run on a same-calendar window one year before each reform with a fake cutover. This gives a *placebo posterior*: what BSTS attributes to that date in a year with no reform. Two cases:

- *Real significant, placebo near zero*: the effect is reform-attributable.

- *Real and placebo both significant and similar in magnitude*: BSTS is picking up a recurring calendar shift its seasonality model does not fully absorb, and the "real" estimate is not separable from that shift.

A sharp filter is the *joint 95% credible interval*: treating real and placebo posteriors as independent, the difference $\bar{\tau}_{\text{real}} - \bar{\tau}_{\text{placebo}}$ has variance $\sigma^2_{\text{real}} + \sigma^2_{\text{placebo}}$, and a 95% interval excluding zero rejects the recurring-calendar story. Reform attribution requires (i) a significant real-window estimate *and* (ii) the placebo not running in the same direction with similar magnitude; the joint interval is reported as a check.

Drift diagnostic: near-window vs. full-window estimate. A characteristic signature of BSTS counterfactual error is an effect that emerges only in the late segment of the post-window: the observed series tracks the counterfactual closely for the first weeks after the cutover and then diverges as the pre-trained extrapolation drifts away from the realised series. We diagnose this by re-running BSTS with a near-cutover post-window (14 days instead of 40); a genuine reform response is similar in sign and magnitude at both window lengths, while drift collapses the near-window estimate toward zero.

3.B. Spec B: per-hour-class BSTS on bid-level outcomes

To identify within-day reallocation in bid *levels*, the same Spec A BSTS is run *independently* on each (tech, market, hour-class) cell for two bid-level outcomes:

- $\bar{p}_{t,m,h,d}$: MW-weighted mean of in-band tranche prices.
- $m_{q;t,m,h,d}$: total in-band quantity, summed across tranches/periods/units (and for IDA also sessions).

The readable summary is the real-window posterior *and the same-calendar 2024 placebo posterior reported side by side for each hour-class separately* (Tables), not the (critical − flat) differential, which would collapse information when two hour-classes move in the same direction at different magnitudes. For each (tech, market, outcome, hour-class) we form $\bar{\tau}_{\text{real}} - \bar{\tau}_{\text{placebo}}$ from two BSTS posteriors (real and placebo) that are independent across years, and use the same joint 95% credible-interval reading as in § 3.A. We avoid per-(single-hour) BSTS because at that resolution the residual noise dominates the seasonal signal.

3.C. Spec C: per-curve DiD for bid shape

Spec B identifies the within-quarter shift in bid *levels*. For identifying bid shapes, we use per tech and bid-shape variables $Y \in \{\sigma_p, N_{\text{eff}}\}$:

$$Y_{u,d,p[,s]} = \alpha_u + \beta\,\mathbf{1}\{d \geq T_r\} + \xi\,\mathbf{1}\{h \in \mathcal{C}\} + \theta\, D_{d,h} + \varepsilon, \qquad \text{SEs clustered by date.}$$

We use DiD rather than BSTS here because the bid-shape outcomes are MCP-centered and MW-weighted (§ 1), so **they do not inherit price-level seasonality**. Differencing absorbs any common-hour seasonality, and under parallel trends on the (critical − flat) differential, θ identifies the ATT on critical-hour bid shape.

I avoid DiD for other level outcomes because of seasonality. I could run DiD on seasonally adjusted variables but we would have to extrapolate the seasonal adjustment from other windows, which would be noisy and potentially biased[9].

Note that flat hours act as a **low-exposure, not perfect, within-day control**: MTU15 is also adopted for flat hours but residual demand small variance does not give incentives to exploit it as in critical hours § 2).

Since we are studying each MTU15 reform separately, we have the standard 2×2 DiD without staggered-treatment negative weights (de Chaisemartin and D'Haultfœ uille, 2026, ch. 3).

3.D. Ongoing extension: BSTS-style counterfactual for the full bid curve

A natural generalisation of Spec B is to replace the scalar in-band summaries ($\bar{p}$, in-band energy) with the *full submitted-bid step function* — treat each daily aggregate supply curve as a functional observation and predict its post-reform counterfactual in the same BSTS spirit.

The motivation is methodological. Modelling seasonality on the full curve is more natural than modelling it on the scalar in-band summaries: the in-band summaries are MCP-relative (the band slides with MCP, and MCP itself has strong seasonality), whereas the supply curve $S(p)$ is defined on the same price domain every day, so weather and fundamentals can be regressed pointwise on p without the band moving underneath the outcome.

We are exploring this via functional factor regression (Otto and Winter, 2025), which controls for seasonality through hourly weather (wind, solar) as functional regressors plus daily gas as a scalar regressor — the same identifying logic as Spec A but on the entire curve rather than on a scalar outcome. The output is a curve-level placebo-net shift $\Delta S(p)$: the gap, at each bid price p, between the post-reform supply curve and the counterfactual the model would have predicted absent the reform. Results are preliminary and not reported here.

4. Results: bid-shape widening, day-ahead CCGT auction-cleared scale-up, and directional price movements

Findings are organised one spec at a time. All three specs share the same wind+solar+gas covariates and the 2024 same-calendar placebo.

[9] Recall that BSTS approach flexibly captures the seasonality without having to adjust for it beforehand.

4.A. Spec A: prices and per-tech cleared energy

		ID15			DA15		
Outcome	**Tech**	Real	Placebo	Net	Real	Placebo	Net
DA clearing price (EUR/MWh)	—	-38.39^{***}	-13.71	-24.68	2.97	-8.54	11.51
IDA clearing price (EUR/MWh)	—	-38.34^{***}	-18.08^{*}	-20.26	5.13	-6.55	11.68
DA cleared (GWh/day)	CCGT	10.48	-10.83	21.31	40.31^{***}	-18.69^{***}	58.99^{***}
	Hydro	-4.36	15.71^{***}	-20.07^{**}	6.16^{*}	9.28^{***}	-3.12
	Hydro pump	-0.82	-0.30	-0.52	4.43^{***}	1.42^{*}	3.02^{*}
	Wind	9.64^{*}	9.54^{*}	0.10	13.16^{***}	2.15	11.00^{*}
	Solar	11.28^{***}	10.53^{***}	0.74	-3.37	1.38	-4.74
	Nuclear	53.69^{**}	-50.45	104.14^{*}	-18.21^{**}	-34.77^{***}	16.56
IDA cleared (GWh/day)	CCGT	-8.08^{***}	-9.18^{***}	1.10	0.68	-2.38	3.06
	Hydro	-1.27^{***}	4.98^{***}	-6.25^{***}	-0.41	1.20^{*}	-1.60^{*}
	Hydro pump	-0.37^{*}	0.54^{**}	-0.91^{**}	0.10	-0.27	0.37
	Wind	5.34^{***}	7.99^{***}	-2.65^{**}	-2.63^{***}	2.28^{*}	-4.91^{***}
	Solar	6.70^{***}	2.55^{***}	4.16^{***}	-5.86^{***}	-0.34	-5.52^{***}
	Nuclear	11.51^{***}	19.06^{***}	-7.56	1.68	-5.43	7.11

Table 3: Spec A BSTS effects on clearing prices and per-tech cleared energy. Bayesian posterior point estimates; stars in Real/Placebo cells are tail-probability stars on each individual posterior ($^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$). The Net column is the placebo-net effect (Real − Placebo) with joint z-test stars derived under independent-Gaussian-posterior approximation: $\text{Net}/\sqrt{\text{SE}^2_{\text{real}} + \text{SE}^2_{\text{placebo}}}$. Windows in § 3.A; placebo columns re-run BSTS on the same calendar window in 2024 with a fake cutover. Observed-vs-counterfactual series in Figures 4–5.

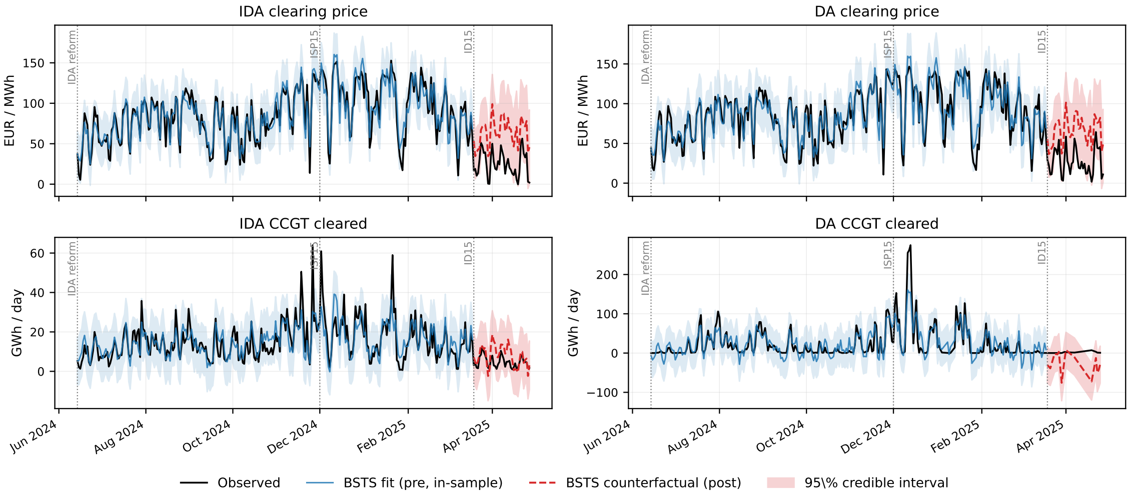

Figure 4: ID15 effects on both markets (2×2 zoomed view, 40-day pre-blackout post-window). Black: observed (7-day MA). Blue: BSTS in-sample fit on the pre-window with 95% band. Red dashed: out-of-sample counterfactual with 95% band. Cutover 2025-03-19; blackout 2025-04-28. Both IDA and DA clearing prices (top row) drop below the counterfactual by similar magnitudes on the same day, even though ID15 only changed intraday granularity — the cross-market rational-anticipation channel into day-ahead (finding (ii)).

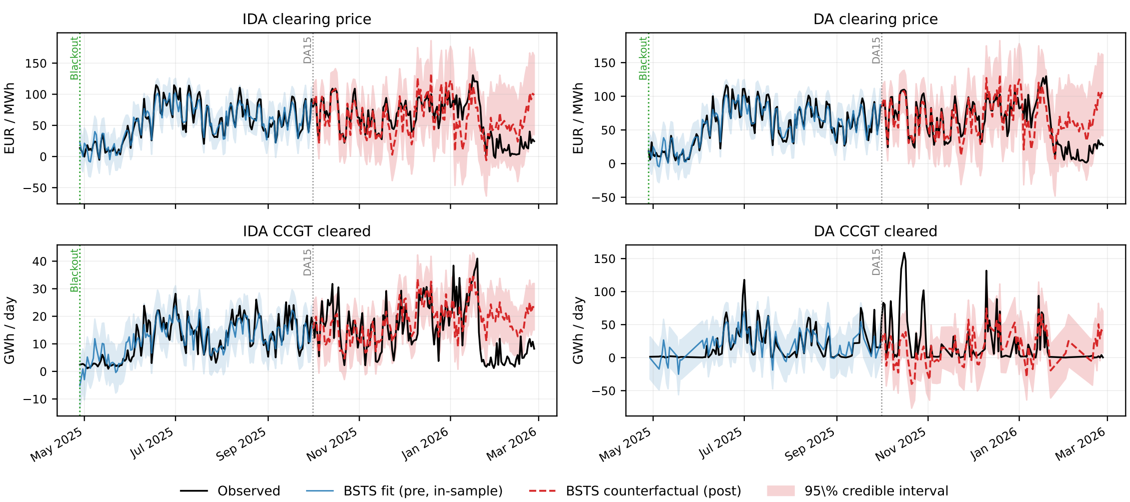

Figure 5: DA15 effects on both markets (2×2 zoomed view, post-window through 2026-02-26). Same colour scheme as Figure 4. Cutover 2025-10-01; reforzada-constant pre-window starts 2025-04-28. The DA CCGT panel (bottom-right) is the one quantity outcome where the observed series visibly departs from the counterfactual post-cutover — the CCGT scale-up of § 4.A.2. By contrast, the IDA panels show no backwards transmission (intraday clearing follows day-ahead, so a DA-side reform cannot propagate backwards).

4.A.1. ID15 clearing-price drop in both markets

Both DA and IDA clearing prices drop sharply under ID15 (Table above; top row of Figure 4). The cross-market symmetry is the signature: both markets drop by similar magnitudes on the same day even though ID15 only changed *intraday* granularity — the rational-anticipation channel into day-ahead. The BSTS conditions on wind, solar, and gas, so the drop is net of the merit-order shift; the pre-window in-sample fit is tight. *Verdict.* The 2024 same-calendar placebo also runs negative; the joint placebo-net 95% interval brackets zero, so the level effect is not retained. We report the raw drop and its cross-market signature only. DA15 price effects are small and not directional.

4.A.2. DA15 day-ahead CCGT cleared scale-up concentrates in critical hours

The only level-shift finding to survive the joint 95% placebo-net test (Figure 5, bottom-right). MTU15-DA lets combined-cycle gas discriminate finely within the hour, raising its expected acceptance probability across the four quarters of a peak hour — same channel as the bid-shape widening in § 4.C.1. Running the Spec A BSTS at hour-class resolution on cleared MWh shows the scale-up concentrates almost entirely in critical hours, with a smaller flat-hour contribution and essentially nothing in midday (where solar saturation displaces gas) — the granularity-bites-where-within-hour-residual-demand-varies channel made direct. The scale-up is also one-sided across markets: substantial in DA, essentially unchanged in IDA — CCGT migrates *into* DA without losing its IDA arm in cleared terms. Extending the post-window through winter shrinks but does not flip the estimate. Hydro and nuclear are essentially null.

Note. OMIE auction-cleared MW is pre-restriction by construction: the adjustments REE makes through the technical-restrictions process do not enter in these programs[10]. The scale-up reported here is therefore a shift in the day-ahead *competitive clearing*, not in REE's restriction call (see § 7).

However, **competitive clearing anticipates expected REE's intervention**, so we should flag this[11].

4.A.3. Renewable cleared MW reallocates toward the granular market

Renewables cannot reposition their physical generation — it follows weather — but aggregators can reposition where the generation is *offered*. Under DA15, day-ahead cleared MW rises broadly across hour-classes for wind (aggregators push the entire output into DA), while the intraday side falls. Solar reallocates time-of-day rather than only market-side: DA cleared rises in midday (where solar generates) and falls in critical hours; IDA cleared falls across the day. Under ID15, intraday cleared MW rises for solar while day-ahead is unchanged.

Mechanism. The same upstream-migration logic as the dispatchable fleet in § 4.A.2 (finer offering raises expected acceptance probability), reinforced by a renewable-specific channel: ISP15 (December 2024) moved REE's imbalance settlement to 15-min, so any intra-hour gap between scheduled and realised generation now triggers a 15-min imbalance penalty. Scheduling into a 60-min market locks the unit into a single hourly schedule even when its 15-min forecast varies within the hour; scheduling into the granular (15-min) market lets the unit submit a schedule that tracks its 15-min forecast and minimises the imbalance bill. Wind and solar have the largest within-hour forecast variance among Spanish techs, so they benefit most from the granular-market shift.

[10] Programa Diario Base de Casación (PDBC), for DA; and Programa Intradiario Base de Casación Incremental (PIBCI), for IDA.

[11] As illustratiion, check SNC-DE-175-17 CNMC 2019 resolution, in which Naturgy was sanctioned for undercommiting with gas plants in the DA and wait to being called in the technical-restrictions Fase I up market, which is pay-as-bid and usually more profitable (check the descriptive section for a description of the issue).

4.A.4. ID15 within-day wedge SD jumps in critical hours

The wedge *level* BSTS is null in both reforms. The within-day wedge SD Δ_d^{SD}, by contrast, jumps sharply under ID15 and concentrates in critical hours.[12] Robust to both a 2024 and an independent 2026 same-calendar placebo.

Mechanism. ID15 makes intraday 15-min while day-ahead stays 60-min, so each clock-hour hosts four IDA quarter-prices against one DA price; $\Delta_{d,h}$ moves more where within-hour residual demand actually varies.

Asymmetric persistence. The symmetric reform (DA15, both markets 15-min) does *not* revert the wedge SD: residual candidates are behavioural hysteresis or a renewable-penetration trend the covariates do not fully absorb.

4.A.5. ISP15 wedge SD jump isolates a settlement-only channel

ISP15 changed REE's settlement period from 60 to 15 min on 2024-12-11 but left the OMIE auction grid unchanged — DA and IDA both still hourly. Mechanically the wedge SD should not move. It does (placebo-net +3.92 EUR/MWh, same direction as ID15). The natural reading is that bidders respond to the finer *settlement* reference itself, not only to a finer auction grid. ID15 then stacks both channels (finer settlement *and* finer auction); ISP15 isolates the settlement-only channel.

4.A.6. BSTS extrapolation drift cases

Applying the drift diagnostic of § 3.A to the Table 3 cells: the headline findings of § 4.A.1 (ID15 price drop) and § 4.A.2 (DA15 day-ahead CCGT scale-up) are visible from the first post-cutover days and stable between 14-day and 40-day post-windows. The pump-storage cleared-MW level effects — on both reforms — fail the diagnostic and we do not interpret them.

4.B. Spec B: within-day bid-volume reallocation supporting the cleared-MW findings

Spec B isolates one bid-side primitive at within-day resolution: $E^{\mathrm{band}}_{t,m,d,\mathcal{X}}$ (in-band offered energy per class-hour, § 1), for each (tech, market, hour-class) cell. Tables 4 and 5 report the real-window and 2024 same-calendar placebo effects side by side per cell; the reform-attributable shift is the real-vs-placebo gap. Reported per hour-class separately rather than as the (critical − flat) differential, which would hide the level story.

[12] Wedge SD placebo-net under ID15: 2.16 EUR/MWh (joint 95% $[0.62, 3.70]$) overall; 3.14 in critical hours (z $\approx$ 4.4). DA15 placebo-net is null.

Tech	Hour-class	DA			IDA		
		Real	Placebo	Net	Real	Placebo	Net
CCGT	morning ramp	-593^{**}	-670^{***}	77	-603^{***}	-2693^{***}	2090^{***}
	midday	-2092^{***}	-594^{***}	-1497^{***}	-1006^{***}	-2432^{***}	1426^{***}
	evening ramp	-166	-357	191	-623^{***}	-2481^{***}	1859^{***}
	flat	-1227^{***}	-364	-863^{**}	-1210^{***}	-2615^{***}	1405^{***}
Hydro	morning ramp	1685^{***}	2755^{***}	-1070^{**}	-632^{**}	2265^{***}	-2898^{***}
	midday	2304^{***}	1476^{***}	828	43	5691^{***}	-5648^{***}
	evening ramp	1903^{**}	3156^{***}	-1253	-287	5023^{***}	-5310^{***}
	flat	2312^{***}	1984^{***}	328	-880^{***}	1545^{**}	-2424^{***}
Hydro pump	morning ramp	-291^{***}	-237^{***}	-54	-206^{**}	-622^{**}	416
	midday	-15	-166^{*}	151	-365^{**}	-1344^{***}	979^{**}
	evening ramp	-270^{***}	-269^{***}	-1	-346^{***}	-1121^{***}	775^{*}
	flat	-205^{***}	-95	-110	-226^{**}	-237	12

Table 4: ID15 Spec B: in-band offered energy (MWh per class-hour, BSTS) by tech, market and hour-class. Real/Placebo cells: BSTS posterior point estimates from the real reform window and the same calendar in 2024 (fake cutover); tail-probability stars $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$ on each individual posterior. Net column: placebo-net (Real − Placebo) with joint z-test stars (independent-posterior approximation).

Tech	Hour-class	DA			IDA		
		Real	Placebo	Net	Real	Placebo	Net
CCGT	morning ramp	1583	-1879^{***}	3462^{**}	-334^{*}	1292^{***}	-1627^{***}
	midday	540^{**}	-448^{*}	988^{**}	1046^{***}	273^{*}	773^{*}
	evening ramp	2851^{***}	392^{**}	2458^{***}	126	278	-152
	flat	1296	-1671^{***}	2966^{**}	-719^{**}	438^{*}	-1157^{***}
Hydro	morning ramp	-42	-82	40	-218	2605^{***}	-2822^{***}
	midday	1226^{***}	904^{***}	322	1010^{***}	2660^{**}	-1650^{*}
	evening ramp	901^{***}	627^{**}	274	116	1873^{**}	-1756^{*}
	flat	-242	-55	-187	-171	1294^{***}	-1466^{**}
Hydro pump	morning ramp	115^{*}	25	89	-312^{***}	297	-609^{**}
	midday	380^{***}	-53	433^{**}	505^{***}	-363	868^{**}
	evening ramp	408^{***}	156	251	-134^{***}	274	-408
	flat	206^{**}	3	203	-289^{**}	-252	-37

Table 5: DA15 Spec B: in-band offered energy (MWh per class-hour, BSTS) by tech, market and hour-class. Real/Placebo cells: BSTS posterior point estimates from the real reform window and the same calendar in 2024 (fake cutover); tail-probability stars $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$ on each individual posterior. Net column: placebo-net (Real − Placebo) with joint z-test stars (independent-posterior approximation).

4.B.1. In-band offered energy migrates into the granular market, with morning ramp dominating under DA15

Under DA15 (Table 5), CCGT offers more in-band energy in DA across all four hour-classes than the 2024 placebo would predict, and less in IDA across all hour-classes except midday. The morning ramp shows the largest real-vs-placebo gap, followed by flat and evening ramp; midday is smallest. The granularity instrument lets combined-cycle gas discriminate within the hour where residual demand actually ramps, and the morning ramp is precisely when that residual demand swings hardest (heating demand rising, solar still negligible). This is the bid-side primitive behind the day-ahead CCGT cleared scale-up of § 4.A.2, and the per-class-hour view shows the within-day source: the morning ramp leads, not the evening ramp despite the latter holding the evening peak prices.

Under ID15 the mirror holds (Table 4): CCGT and pump-storage offer more in-band energy in IDA across hour-classes than the placebo would predict, and less in DA. The placebo intercept itself is large and negative for ID15 IDA — a recurring cross-year shift the reform reading sits on top of — so the reform-attributable shift is the gap (real − placebo), not the absolute real-window value. The migration is broad-based on the dispatchable side, not concentrated in any one sub-class.

Pump-storage in midday under both reforms. Pump-storage shows a midday-specific expansion in DA under DA15 and in IDA under ID15 — the within-day arbitrage value of pump-storage peaks at the solar-saturated midday window, and the granularity instrument lets that arbitrage be priced more finely.

Hydro is the exception. Under both reforms, hydro pulls in-band offered energy *out* of IDA across hour-classes — the opposite of the dispatchable migration story. The most plausible reading is that the 2024 same-calendar placebo is a poor benchmark for hydro: reservoir filling and inflows differ substantially across years, and Spec A's wind+solar+gas covariates do not absorb this hydro-specific seasonal variation. We do not lean on hydro for the migration story.

Bid vs. clearing caveat. Bidding migrates into the granular market; clearing follows only for DA15 CCGT. ID15 CCGT cleared MW (Spec A) is essentially unchanged in IDA — the in-band offered energy expansion does not translate to a cleared-MW expansion at the IDA clearing price.

The portfolio migration of wind and solar (§ 4.A.3) is invisible in this outcome — renewables bid mostly as single-price blocks at low EUR/MWh outside the $\pm h$ competing tier — but it shows directly in the cleared MW per market (Table 3).

4.C. Spec C: bid-shape findings supporting the price, scale-up and wedge-SD claims

Spec C identifies within-quarter bid-*shape* effects: how the bid ladder around MCP changes across the four quarters of a critical hour. Two outcomes: σ_p (within-quarter SD of in-band bid prices) and N_{eff} (effective tranche count, 1 for a single block, large for many equal tranches). Tech scope: CCGT (the focal scarcity-tier technology), hydro and pump-storage (dispatchable comparison), wind (price-taking falsifier).[13] Solar excluded (no in-band scarcity-tier capacity); nuclear excluded (mostly bilateral). The three subsubsections below match the three § 4.A findings these bid-shape outcomes most directly speak to.

[13] Wind has the dominant sell-side quantity but is not the marginal price-setter; CNMC (2025c) defines the marginal technologies as coal + CCGT + hydro + 5% RECORE. Aggregator portfolios construct dense single-price blocks at low EUR/MWh, so EUPHEMIA partially accepts a wind unit as the mechanical residual — not strategic price-setting.

Outcome (DiD θ)	Subsample	ID15 DA (h=50)	ID15 IDA (h=62)	DA15 DA (h=50)	DA15 IDA (h=58)
σ_p (EUR/MWh)	CCGT	4.17^{***} (0.44)	0.10 (0.53)	0.68^{***} (0.22)	-0.68^{*} (0.36)
	Hydro	-0.53^{*} (0.29)	0.43^{*} (0.25)	1.80^{***} (0.34)	0.30 (0.25)
	Hydro pump	-0.73^{***} (0.27)	−0.01 (0.18)	0.92^{***} (0.35)	0.32^{*} (0.19)
	Wind†	-0.10^{***} (0.03)	0.04^{**} (0.02)	−0.02 (0.03)	-0.02^{***} (0.003)
N_{eff}	CCGT	−0.08 (0.21)	0.79 (0.73)	1.89^{***} (0.15)	−0.17 (0.53)
	Hydro	-0.24^{***} (0.06)	0.18^{***} (0.05)	0.51^{***} (0.09)	0.00 (0.06)
	Hydro pump	0.07 (0.05)	0.12^{*} (0.07)	−0.06 (0.07)	−0.16 (0.13)
	Wind†	0.01 (0.02)	0.03^{***} (0.01)	−0.01 (0.01)	−0.01 (0.01)

Table 6: Spec C per-curve DiD on bid-shape outcomes. SEs in parentheses, clustered by date. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Per-(tech, market) subsample regressions, unit FE absorbed via within-unit demeaning. Bandwidth h is window-and-market-specific (the p90 of the MCP distribution in that window, market by market): see § 1 and § 3.C. Sensitivity to wider bandwidths is reported in § 5(e). †Wind reads as a price-taking baseline: 84% of in-band wind sell curves are a single block; magnitudes are an order of magnitude smaller than the dispatchable techs.

4.C.1. DA15 widens day-ahead CCGT bid-shape in critical hours — the bid-side primitive of the cleared scale-up

Under DA15, day-ahead CCGT σ_p and N_{eff} widen substantially in critical hours (Table; Figure 6): combined-cycle gas submits four distinct quarter-prices per hour, exploiting within-hour residual-demand variation that previously had to be priced at a single hourly value. The finer within-quarter price discrimination raises the expected acceptance probability of CCGT MW across the four quarters of a peak hour — the bid-side mechanism behind the DA15 day-ahead CCGT cleared scale-up in critical hours (§ 4.A.2). Hydro and pump-storage widen their DA σ_p on the same window with the same sign, smaller magnitude.

4.C.2. ID15 widens IDA dispatchable bid ladders — the bid-side primitive of the IDA price drop and wedge-SD jump

Under ID15, IDA N_{eff} widens significantly for hydro, pump-storage, and wind — more granular ladders in the now-15-min IDA. The dispatchable fleet (and the wind price-taker) exploits the within-hour residual-demand variation that becomes priceable. Two consequences in § 4.A:

- *ID15 price drop* (§ 4.A.1): more granular IDA ladders intensify within-quarter competition at the competing-tier prices that set MCP, lowering the IDA clearing price.
- *ID15 wedge-SD jump in critical hours* (§ 4.A.4): with IDA quarter-prices diverging across the four quarters of a critical hour while DA remains a single hourly price, the per-clock-hour DA − IDA wedge moves more across the 24 clock-hours.

CCGT N_{eff} in IDA does not widen significantly under ID15 — combined-cycle gas competes primarily in DA at this stage.

4.C.3. Cross-market substitution: DA15 tightens CCGT IDA bid-shape; ID15 widens CCGT DA bid-shape

The reform that lands the granularity instrument on one market sets off a complementary bid-shape adjustment on the other:

- Under ID15, CCGT widens its DA σ_p substantially even though only IDA became granular — combined-cycle gas restructures its DA ladder *in anticipation* of finer IDA re-trading. This is the cross-market spillover behind the DA-side ID15 price drop in § 4.A.1: DA prices fall on the same day as IDA prices because DA bidders adjust to the new IDA-anchored expectation.
- Under DA15, CCGT IDA σ_p *tightens*: combined-cycle gas pulls its structured ladder out of IDA *into* DA in critical hours, consistent with the one-sided DA-only cleared scale-up in § 4.A.2.

Wind reads as a price-taking baseline throughout.

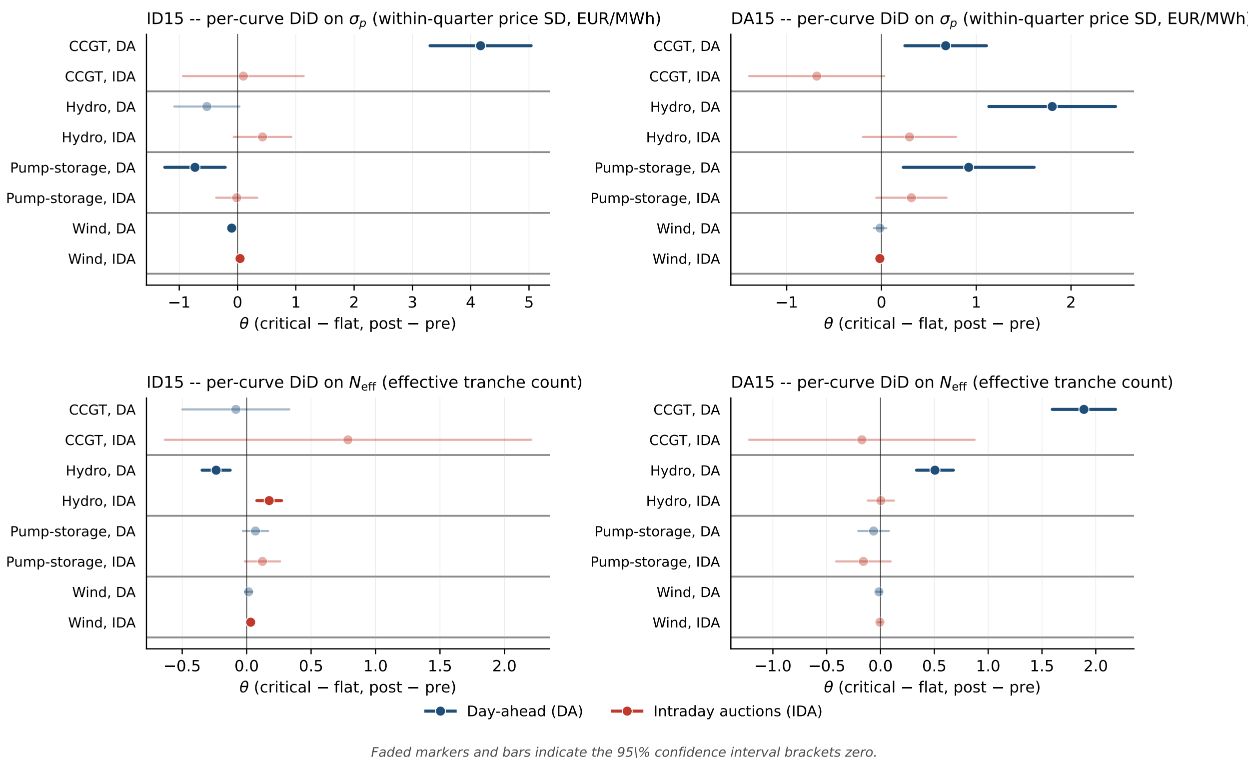

Figure 6: Spec C per-curve DiD on bid-shape outcomes at the window-specific p90 bandwidth. *Top row:* σ_p DiD (within-quarter price SD); *Bottom row:* N_{eff} DiD (effective tranche count). *Left column:* ID15. *Right column:* DA15. Each row inside a panel is one (tech, market) cell; blue = day-ahead, red = intraday auctions. Faded markers indicate the 95% CI brackets zero. Bandwidth h varies by (window, market): ID15 uses h=50 (DA) / h=62 (IDA); DA15 uses h=50 (DA) / h=58 (IDA). The picture is genuinely market-specific: ID15's effects concentrate in IDA N_{eff} (more granular intraday ladders) and DA CCGT σ_p (anticipation-channel widening); DA15's effects concentrate in DA across all dispatchables, with CCGT IDA σ_p tightening (substitution into DA).

5. Robustness

5.1. Pre-trend placebo — the headline check

A pre-only midpoint placebo splits each reform's pre-window at its midpoint and runs a fake critical-flat DiD on the same outcome. ID15 IDA dispatchable σ_p placebos all bracket zero — the ID15 bid-shape findings rest on a clean pre-trend. For DA15 day-ahead: hydro σ_p placebo is clean, CCGT σ_p placebo runs opposite the headline (so de-trending strengthens it), and pump-storage σ_p's placebo exceeds the headline DiD — not separable from a pre-existing trend.

5.2. Hour-class falsification (midday vs. flat)

A midday-vs-flat falsification gives sign-stable but smaller-magnitude DiDs at p_{90}: the reform reshapes the entire non-flat zone, consistent with § 4.B. The cleaner critical-vs-flat contrast remains the headline because flat hours are the unambiguous within-day control.

5.3. Cross-border and reforzada — the two natural confounds

SIDC/XBID also moved to 15-min on the same date as ID15 — a sibling treatment on the same reform, not an OVB: adding hourly cross-border net flow as a control on ID15 IDA σ_p leaves the per-tech DiDs essentially unchanged. The reforzada regulatory stance acts on the *level* of Fase I up-redispatch, not its *within-hour SD*: reforzada parks capacity in the scarcity tier and recalls it through restriction offers (§ 7), while MTU15 reshapes the competing tier within the band — two non-overlapping mechanisms.

5.4. Per-CCGT-firm decomposition of the DA15 σ_p widening

The pooled DA15 CCGT day-ahead σ_p widening is driven primarily by Naturgy and to a lesser extent Iberdrola; Endesa and EDP-HC do not contribute. The Naturgy effect is not a flat-baseline artefact: σ_p rises in both arms but more in critical hours.

5.5. Bandwidth h

p_{90} isolates the competing tier and excludes the parked tier above the ceiling (a non-strategic placeholder, § 7). Including the parked tier would bias σ_p upward by a measurement-driven cross-tier spread. The picture is robust to small shifts around p_{90}; at much wider bandwidths, CCGT σ_p magnitudes inflate and several DiD coefficients flip sign because the cross-tier spread moves across the cutover for reasons unrelated to within-tier bidding. The IDA bandwidth pools the three sessions; session 3 has a materially wider $p_{90} - p_{50}$ than sessions 1–2, but a session-specific h cannot flip signs — the per-session BSTS in § 5.6 shows the price-level effect is uniform across sessions.

5.6. Per-IDA-session BSTS

Running Spec A separately on each IDA session's daily mean price (Table 7) confirms session symmetry: the ID15 placebo-net drop is essentially identical across S1, S2, S3, and the DA15 IDA-side rise sits in a narrow positive band. ID15 placebo uses the 2026 same-calendar window (the 2024 placebo predates the 6→3 IDA reform and is not session-comparable).

Reform	Session	Effect (real)	Placebo	Placebo-net
ID15 IDA	S1	-79.8^{***}	-16.1^{*}	-63.7^{***}
	S2	-79.9^{***}	-11.4	-68.5^{***}
	S3	-81.9^{***}	-8.4	-73.5^{***}
DA15 IDA	S1	15.4^{***}	3.9	11.5^{**}
	S2	14.4^{***}	-1.8	16.2^{***}
	S3	19.0^{***}	10.6^{**}	8.4

Table 7: Per-IDA-session BSTS on daily clearing prices. BSTS posterior means in EUR/MWh; tail-probability stars $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. ID15 placebo is the 2026 same-calendar window (3 sessions, matching real). DA15 placebo is 2024 same-calendar (3 sessions). Same windows and covariates as Spec A. Session symmetry implies pooling across sessions in the headline IDA BSTS is innocuous: a session-specific bandwidth would not change the level conclusion.

5.7. Pump-storage seasonal control

The DA15 day-ahead pump-storage rise is borderline at joint 95% in the main BSTS because of cross-season baseline drift between summer pre and autumn post. A same-calendar DiD restricted to Oct–Dec 2024 vs. Oct–Dec 2025 and a Fourier deseasonalisation both retain the directional finding.

6. Continuous-market response at the three structural reforms

Specification. BSTS-style counterfactual on daily ES-leg continuous-market series (one of `seller_zone` or `buyer_zone` = `10YES-REE------0`), with wind+solar+gas covariates and a weekly seasonal cycle (same setup as § 3.A). *Units.* All quantity outcomes are *energy* (MWh = MW × delivery hours = `quantity_mw` × `mtu_minutes`/60, summed per day), not raw MW sums: cross-reform comparability requires the stock (energy delivered), not the rate, since post-MTU15 each clock-hour holds four 15-min contracts where pre-MTU15 it held one 60-min contract. The volume-weighted mean price is energy-weighted (weight = MWh of each trade). Three reform cutovers: *ISP15* (settlement period 60 → 15 min, 2024-12-11; pre 2024-06-14 → 2024-12-10 holds the IDA reform constant, post 2024-12-11 → 2025-03-18 stops at ID15); *ID15* (intraday auctions + continuous market → MTU15, 2025-03-19; pre 2024-12-11 → 2025-03-18 post-ISP15, post 2025-03-19 → 2025-04-27 pre-blackout); *DA15* (day-ahead → MTU15, 2025-10-01; pre 2025-04-28 → 2025-09-30 post-blackout, post 2025-10-01 → 2025-12-31). Each real arm has a 2023/2024 same-calendar placebo with a fake cutover.

Reform	Outcome	Real	Placebo	Placebo-net
	n_{trades}/day	$-3{,}120^{***}$	$-1{,}231^{***}$	$-1{,}889$
ISP15	GWh/day	-3.85^{*}	-2.23^{**}	-1.62
	VW-price (EUR/MWh)	-15.1	-27.4^{***}	12.3
	n_{trades}/day	$22{,}335^{***}$	$1{,}381^{*}$	$20{,}954^{***}$
ID15	GWh/day	-7.19^{***}	6.72^{***}	-13.91^{***}
	VW-price (EUR/MWh)	-80.1^{***}	-14.1^{***}	-66.0^{***}
	n_{trades}/day	$5{,}018^{**}$	257	4,761
DA15	GWh/day	3.85^{***}	-1.35	5.20^{**}
	VW-price (EUR/MWh)	4.7	-4.7	9.4

Table 8: Continuous-market response at the three structural reforms. BSTS posterior means; tail-probability stars $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. Placebo-net joint significance based on independent posteriors propagating two variances.

(vi) ISP15: settlement scope, no contract-structure response. The December 2024 reform changed how REE settles imbalance positions (60→15-min ISP), but it did not change the OMIE continuous-market contract grid: traders still bought and sold hourly delivery contracts before and after. A null effect is therefore the correct prediction. Trade count drops 22% raw and energy turnover drops 14% raw, but the same drift is in the 2023 placebo (settlement bookkeeping innovations are not the source) and all three placebo-net intervals bracket zero. We report this as a methodological sanity check: a settlement-only reform shows nothing on contract-structure outcomes, consistent with our specification's reading of where the reform lives.

(vii) ID15: the contract grid moves from 60 to 15 minutes — the structural reform. Pre-reform, each clock-hour of delivery hosted a single 60-min continuous-market contract; post-reform, four separate 15-min contracts. Three of our four outcomes pop accordingly:

- *Trade count.* Each hour now offers four matching opportunities where there used to be one, so the trade-count surge is mechanical at first order (placebo-net well clear of zero). Even "mechanical" is informative — it confirms market participants *used* the new finer grid rather than continuing to trade only on the hourly aggregate.
- *Energy turnover.* The placebo-net is *negative*: more contracts, but each smaller, and the total energy changing hands drops. Behavioural read: the granularity reform did not pull additional energy into the continuous market; it sliced the same (in fact slightly less) energy into smaller pieces. The drop is partly because some hourly counterparties drop out when they don't see a single 60-min match on offer, and partly because dispatchables migrated bidding into the now-finer IDA auctions instead (§ 4(v)).
- *VW-price.* The continuous market clears in equilibrium with the auctions, and we see the same ID15 day-of-cutover clearing-price drop here as on auction-cleared prices, well clear of zero — the cross-market spillover documented for ID15 in § 4(ii).

(viii) DA15: a smaller, indirect continuous-market effect. Day-ahead granularity does not change the continuous-market contract grid (already 15-min post-ID15); only the day-ahead *reference price* that continuous traders watch becomes finer. With a finer DA reference, traders reposition more aggressively in continuous: daily GWh up (joint 95% retains), trade count borderline. The price effect is statistical noise. The interpretation is that DA15's continuous-market footprint is incremental and proportional to how much tighter the day-ahead reference now is, not a structural break — exactly as expected from a reform that touches the reference market but not the continuous-market contract architecture.

7. Two market-structure facts that frame the bidding evidence

The two-tier CCGT bid curve and the day-ahead/intraday contrast. Every Big-4 CCGT fleet runs a two-tier day-ahead offer: a low *competing* tier near marginal cost and a high *parked* tier above the clearing-price ceiling. Withholding is firm-wide, not a Naturgy story. The same fleets bid a single competitive tier with negligible withholding in the intraday auction, after Fase I has cleared. A unit's marginal cost does not change between two auctions hours apart. The parked tier is a placeholder rather than a Fase I bid: Fase I up-redispatch is settled through a separate restriction offer submitted to REE.[14] MTU15-DA grants CCGT a within-hour degree of freedom over its competing tier; the CCGT auction-cleared scale-up in § 4(iii) is the competing tier expanding into the four quarters of each peak hour.

Market concentration migrated from wholesale to ancillary services. Wholesale-market HHI hovers around 1 100 over 2016–2024 (below the moderate-concentration threshold of 1 500) and falls mid-day when solar peaks (solar is less concentrated than the Big-3). Day-ahead restrictions upward HHI above 4 000; deviations upward above 6 000; tertiary upward around 5 000. Median of 4 firms with accepted upward restrictions bids. As renewables eroded wholesale-market power, technical and locational requirements concentrated the ancillary markets that absorb their variability.

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[14] Procedure PO 3.2, settlement PO 14.4 ap. 20.1; see also CNMC (2025b,a). Earlier drafts of this memo conflated the two; the distinction justifies measuring bid shape and bid levels in the competing tier only, which is what the window-specific p_{90} bandwidth in § 1 achieves.

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